

Heart Disease Prediction

Project Overview & README

Machine Learning | Python | Streamlit

Academic Project Documentation

Version 1.0 | 2025

1. Project Overview

Heart Disease Prediction is a machine-learning-powered web application that predicts whether a patient is likely to have heart disease based on clinical parameters such as age, blood pressure, cholesterol, and more. The application is built with Python and Streamlit, making it accessible directly through a browser with no installation effort for end-users.

2. Features

- Interactive web interface powered by Streamlit
- Real-time heart-disease prediction using a trained Random Forest classifier
- Input validation with user-friendly error messages
- Prediction probability display with risk classification
- Clean, responsive UI suitable for demonstrations and college submissions
- Modular codebase — easy to extend with new models or features

3. Technologies Used

Layer	Technology	Version
Language	Python	3.10+
Web Framework	Streamlit	1.30+
ML Library	Scikit-learn	1.4+
Data Processing	Pandas / NumPy	2.x / 1.x
Model Persistence	Joblib	1.3+
Visualisation	Matplotlib	3.8+

4. Project Structure

```
Heart-Disease-Prediction/
■■■ app.py # Streamlit application entry point
■■■ train_model.py # Model training script
■■■ prediction.py # Prediction helper module
■■■ model.pkl # Serialised trained model
■■■ requirements.txt # Python dependencies
■■■ README.md # Project readme
■■■ .gitignore
■■■ LICENSE
■■■ dataset/
■ ■■■ heart.csv # UCI Heart Disease dataset
■■■ screenshots/
■ ■■■ home.png
■ ■■■ input.png
■ ■■■ result.png
■■■ docs/
■ ■■■ README.pdf
■ ■■■ SRS.pdf
■ ■■■ System_Design.pdf
■ ■■■ Project_Report.pdf
■■■ models/
■■■ model.pkl
```

5. Dataset Information

The project uses the UCI Heart Disease Dataset (Cleveland subset), which is publicly available on Kaggle and the UCI Machine Learning Repository.

Feature	Type	Description
age	Integer	Age of the patient (years)
sex	Binary	1 = Male, 0 = Female
cp	Categorical (0-3)	Chest pain type
trestbps	Integer	Resting blood pressure (mm Hg)
chol	Integer	Serum cholesterol (mg/dl)
fbs	Binary	Fasting blood sugar > 120 mg/dl
restecg	Categorical (0-2)	Resting ECG results
thalach	Integer	Maximum heart rate achieved

Feature	Type	Description
exang	Binary	Exercise-induced angina
oldpeak	Float	ST depression induced by exercise
slope	Categorical (0-2)	Slope of peak exercise ST
ca	Integer (0-4)	Number of major vessels coloured by fluoroscopy
thal	Categorical	Thalassemia type
target	Binary	1 = Heart Disease, 0 = No Disease

6. Installation & Usage

Prerequisites

- Python 3.10 or above
- pip package manager
- Git (optional)

Steps

```
# 1. Clone the repository
git clone https://github.com/yourusername/Heart-Disease-Prediction.git
cd Heart-Disease-Prediction

# 2. Install dependencies
pip install -r requirements.txt

# 3. Train the model (optional - model.pkl already included)
python train_model.py

# 4. Launch the application
streamlit run app.py
```

After running the last command, open your browser at <http://localhost:8501>

7. Future Enhancements

- User authentication and prediction history dashboard
- PDF report download for each prediction
- Doctor recommendation system
- REST API (FastAPI) for third-party integrations
- Cloud deployment on Heroku / AWS / GCP
- Mobile-responsive redesign

- SHAP / LIME explainability charts
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8. License & Author

This project is licensed under the MIT License.

Developed as part of an academic final-year / mini project submission. All dataset credits go to the UCI Machine Learning Repository.